

PROJECT DESIGN PHASE-II

Solution Specifications (Functional & Non-Functional)

Submission Date	31 January 2025
Team ID	
Project Name	ShopEz
Student Details	Alladi Manjunadh Computer Science Engineering alladimanjunadh07@gmail.com
Assessed Marks	4 Marks

Functional Requirements (FR)

These components outline the specific operations and capabilities the ShopEz application must provide to the end-user.

FR No.	Requirement Domain (Epic)	Technical Deliverables (Stories / Tasks)
FR-1	User Account Lifecycle	- Data-driven registration via standard web forms - Seamless authentication using Google/Gmail integrations - Secure sign-up through professional LinkedIn providers
FR-2	Identity Verification	- Automated account validation through email links - Multi-factor confirmation using SMS-based One-Time Passcodes
FR-3	Product & Cart Logic	- Synchronous cart updates with MongoDB persistence - Dynamic catalog filtering and categorized sorting
FR-4	Order Fulfillment	- Checkout processing with integrated secure gateways - Generation of digital receipts and persistent order history

Non-Functional Requirements (NFR)

These attributes define the quality standards and operational constraints for the ShopEz architecture.

NFR No.	Quality Attribute	Strategic Objective / Description
NFR-1	Usability	

NFR No.	Quality Attribute	Strategic Objective / Description
		Ensuring a responsive UI/UX across mobile and desktop utilizing React.js components.
NFR-2	Security	Implementing JWT-based session management and password hashing (Bcrypt) for data safety.
NFR-3	Reliability	Guaranteeing system consistency and data integrity through robust MongoDB transactions.
NFR-4	Performance	Optimizing API latency to ensure page sub-second response times during peak catalog load.
NFR-5	Availability	Maintaining continuous service access through managed cloud infrastructure (MongoDB Atlas).
NFR-6	Scalability	Supporting high concurrent user traffic via the stateless nature of Node.js/Express.